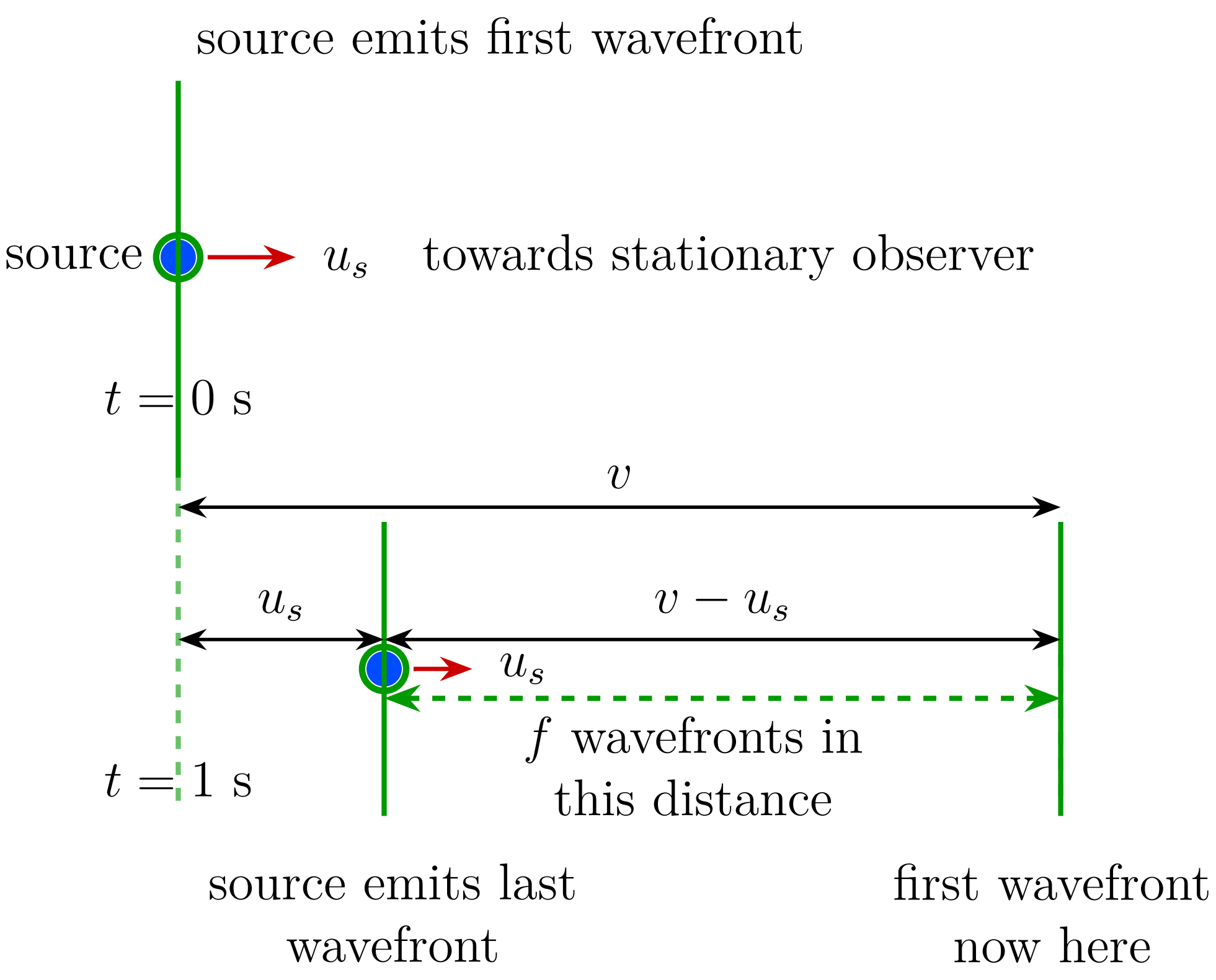
source emits first wavefront
source
u_s
towards stationary observer
$t = 0$ s
v
u_s
$v - u_s$
u_s
f wavefronts in
this distance
$t = 1$ s
source emits last
wavefront
first wavefront
now here